Dr. Agughasi Victor Ikechukwu

Senior Assistant Professor | AI for Healthcare | Medical Imaging | Explainable AI | Multimodal and Agentic AI

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Professional Summary

Senior Assistant Professor in the Department of Computer Science and Medical Engineering, School of Engineering, Dayananda Sagar University, Bangalore, with a Ph.D. in Computer Science and experience in artificial intelligence, medical imaging, explainable AI, multimodal machine learning, computer vision, deep learning, clinical decision support, assistive technology, and AI-enabled healthcare systems. Published in Scopus, SCI, SCIE-indexed journals and IEEE conferences; reviewer for Q1/Q2 journals; mentor of award-winning student innovation teams; contributor to competitive AI healthcare grants, patents, and institutional AI funding initiatives.

Core Competencies and Keywords

Artificial Intelligence; Machine Learning; Deep Learning; Computer Vision; Medical Image Analysis; Explainable AI; Multimodal Machine Learning; Agentic AI; Context Engineering; Data Science; Clinical Decision Support; Assistive Technology; Speech and Hearing Technology; Infant Biometrics; Chest X-ray Analysis; COPD Diagnosis; Diabetic Retinopathy; Breast Cancer Thermography; PET-CT Analysis; Radiomics; Python; Java; JavaScript; PHP; MySQL; PostgreSQL; Oracle; Research Methodology; Academic Mentorship; Grant Writing; Patent Development; Sports and Student Leadership.

Selected Achievements, Grants, Innovation and Leadership

- **Winner, AIISH Hackathon 2025 on Assistive Technology:** Project titled "Development of Interactive Teleconsultation Tool for SLPs and Audiologists", focused on strengthening remote support for speech-language pathologists and audiologists.
- **Phase 1 Winner / Final-Round Shortlisted Team, AIISH Hack'A'Comm 2026:** Guided team led by Ms. Afifa Taskeen for "Adaptive Augmentative and Alternative Communication (AAC) for Children with Cerebral Palsy (CP)", proceeding to prototype development.
- **Invited Keynote Speaker, ICRCET 2026:** Invited keynote speaker at the 15th International Conference on Recent Challenges in Engineering and Technology (ICRCET 2026), Bangalore, India, organized by IFERP Academy, 22-23 April 2026.
- **VGST K-FIST Level 2 Proposal - INR 30 Lakhs:** PI for "Development and Clinical Validation of AI-Assisted Newborn Footprint-Palmprint Biometrics for Infant Identity Assurance"; shortlisted / accepted for final evaluation on 2 June 2026.
- **AI Centre of Excellence Proposal - INR 20 Crores:** Institutional proposal shortlisted for final approval to strengthen AI research, innovation, training, and deployment capacity.
- **Patent Publications Awaiting Examination:** Three Indian patent publications in AI / healthcare technology, including Indian Patent Application No. 202641015976, "AI-Assisted Newborn Footprint and Palmprint Recognition for Reliable Infant Identity Verification".
- **Sports Coordinator and Winner - MITM Mahadasara Sports Utsava 2026:** Contributed to sports leadership, coordination, student engagement, and institutional team spirit.

Education

Ph.D. in Computer Science, University of Mysore, Karnataka, India *Sept. 2019 - July 2024*

Thesis: *Machine Learning Algorithm for the Diagnosis of Chronic Obstructive Pulmonary Diseases from Chest X-ray Images.*

M.Sc. in Computer Science, Bangalore University, Bangalore, India *Apr. 2014 - Mar. 2016*

Academic and Industry Experience

Senior Assistant Professor, Department of Computer Science and Medical Engineering *June 01, 2026 - Present*

School of Engineering, Dayananda Sagar University, Bangalore, Karnataka, India

AI for Healthcare, Medical Image Analysis, Machine Learning, Computer Vision, Explainable AI, Multimodal AI, and Research Mentorship.

- Joined DSU to contribute to interdisciplinary teaching, research, innovation, and student mentorship at the interface of computer science, medical engineering, and AI-enabled healthcare.
- Focus areas include medical image analysis, explainable AI, infant biometrics, assistive technology, quantum machine learning for healthcare, and clinical decision-support systems.

Assistant Professor, Department of Computer Science and Engineering (Artificial Intelligence) *Oct. 2021 - May 2026*

Maharaja Institute of Technology Mysore, India

Machine Learning, Computer Vision, Big Data Analytics, Digital Image Processing, DBMS, Python for Data Visualization, Research Methodology.

- Taught and supervised undergraduate students in AI, machine learning, healthcare AI, computer vision, data analytics, and research-oriented project development.
- Guided innovation teams working on assistive technology, AAC systems, EEG analysis, infant biometrics, medical imaging, and applied machine learning systems.
- Contributed to institutional research proposals, hackathons, workshops, sports coordination, student mentoring, and

academic administration.

Research Associate *June 2018 - Oct. 2021*
Maharaja Institute of Technology Mysore, India
Machine Learning, Big Data Analytics, Machine Learning Projects, Python, Mobile App Development in Java.

- Supported teaching and research activities in machine learning, data analytics, Python programming, mobile application development, and student projects.
- Developed applied research experience in medical image analysis, explainable AI, and ML-based clinical decision support.

Visiting Faculty (Voluntary) *Aug. 2018 - May 2019*
Dr. Ambedkar Institute for Management Science, Bangalore, India
Information System and Science, Database Management Systems. **Visiting Faculty (Voluntary)** *Jul. 2017 - Feb. 2018*
St. Aloysius Degree College, Bangalore, India
Information System and Science, Database Management Systems. **Teaching Assistant (Voluntary)** *Oct. 2014 - Mar. 2016*
St. Joseph's College, Bangalore, India
Computer Fundamentals and Web Design using PHP.

Research Interests

Medical Imaging; Explainable AI Models; Data Science; Multimodal Machine Learning; Deep Learning; Computer Vision; Context Engineering; Agentic AI Models; Clinical Decision Support; Assistive Technology; AI-Enabled Healthcare Systems; Infant Identity Assurance; Healthcare Data Analytics; Tele-consultation Systems.

Intellectual Property and Scholarly Outputs

- **Book manuscript awaiting publication:** Completed draft of first Quantum ML book, *Quantum Machine Learning Cookbook for Medical Image Analysis: Recipes, Code Patterns, and Research Pathways for Early-Career Researchers.*
- **Patent publications:** Three Indian patent publications in AI / healthcare technology; status: awaiting examination.
- **AI-Assisted Newborn Footprint and Palmprint Recognition for Reliable Infant Identity Verification**, Indian Patent Application No. 202641015976; filed 13 February 2026; published 20 February 2026; awaiting examination.

Funded and Shortlisted Research / Innovation Proposals

- **VGST K-FIST Level 2 Proposal - INR 30 Lakhs:** "Development and Clinical Validation of AI-Assisted Newborn Footprint-Palmprint Biometrics for Infant Identity Assurance"; accepted / shortlisted for final evaluation scheduled on 2 June 2026.
- **AI Centre of Excellence Proposal - INR 20 Crores:** Institutional proposal shortlisted for the final round of approval to establish AI research, innovation, training, and deployment infrastructure.

Selected Innovation and Hackathon Projects

- **Development of Interactive Tele-consultation Tool for SLPs and Audiologists** – Winner, AIISH Hackathon 2025 on Assistive Technology.
- **Adaptive AAC for Children with Cerebral Palsy** – Phase 1 Winner / Final-Round Shortlisted Team, AIISH Hack'A'Comm 2026; guided team led by Ms. Afifa Taskeen.
- **AI-Assisted Newborn Footprint-Palmprint Biometrics for Infant Identity Assurance** – PI-led healthcare biometrics research proposal for infant identity assurance.
- **AI-Assisted Multi-Task EEG Analysis for Epilepsy and Related EEG-Based Conditions** – Student healthcare AI project involving clinical consultation and neurologist guidance.

Publications

1. "Firefly-Based Segmentation and Residual Deep Learning for Multi-Class Diabetic Retinopathy Detection", *Inteligencia Artificial*, DOI: `10.4114/intartif.vol28iss76pp223-252`.
2. "Enhancing Early Breast Cancer Detection with Infrared Thermography: A Comparative Evaluation of Deep Learning and Machine Learning Models", *Technologies*, DOI: `10.3390/technologies13010007`.
3. "Leveraging Transfer Learning for Efficient Diagnosis of COPD Using CXR Images and Explainable AI Techniques", *Inteligencia Artificial*, DOI: `10.4114/intartif.vol27iss74pp133-151`.
4. "The Superiority of Fine-tuning over Full-training for the Efficient Diagnosis of COPD from CXR Images", *Inteligencia Artificial*, DOI: `10.4114/intartif.vol27iss74pp62-79`.
5. "CX-Net: An Efficient Ensemble Semantic Deep Neural Network for ROI Identification from Chest X-ray Images for COPD Diagnosis", *Machine Learning: Science and Technology*, DOI: `10.1088/2632-2153/acd2a5`.
6. "Energy-Efficient Deep Q-Network (EEDQN): Reinforcement Learning-based Approach to Efficient Routing Protocol in Wireless Internet-of-Things", *Indonesian Journal of Electrical Engineering and Computer Science*, DOI: `10.11591/ijeces.v33.i2.pp971-980`.
7. "COPDNet: An Explainable ResNet50 Model for the Diagnosis of COPD from CXR Images", *IEEE INDISCON*, 2023, DOI: `10.1109/INDISCON58499.2023.10270604`.
8. "xAI: An Explainable AI Model for the Diagnosis of COPD from CXR Images", *IEEE ICDDS*, 2023, DOI: `10.1109/ICDDS59137.2023.10434619`.
9. "i-Net: A Deep CNN Model for White Blood Cancer Segmentation and Classification", *International Journal of Advanced Technology and Engineering Exploration*, DOI: `10.19101/IJATEE.2021.875564`.
10. "ResNet-50 vs VGG-19 vs Training from Scratch: A Comparative Analysis of the Segmentation and Classification of Pneumonia from Chest X-ray Images", *Global Transaction Proceedings*, DOI: `10.1016/j.gltp.2021.08.027`.
11. "Semi-Supervised Labelling of Chest X-Ray Images Using Unsupervised Clustering for Ground-Truth Generation", *Applied*

Engineering and Technology, DOI: `10.31763/aet.v2i3.1143`.

12. "Effective Approach for Fine-Tuning Pre-Trained Models for the Extraction of Texts From Source Codes", *ITM Web of Conferences*, DOI: `10.1051/itmconf/20246503004`.

Conference Papers Presented

- "Evaluating the Feasibility of a Pre-Processing Framework for Enhanced Text Information Extraction from Source Codes", ADCIS-2024, BITS Pilani Goa, India.
- "Advances in Thermal Imaging: A Convolutional Neural Network Approach for Improved Breast Cancer Diagnosis", ICDCOT-2024, SJBIT, India.
- "Optimizing ResNet50 and VGG19 Networks for Accurate COPD Diagnosis Through CXR Analysis", ERCICAM-2024, NITTE Meenakshi Institute of Technology, India.
- "ResNet-50 vs Training from Scratch: A Comparative Analysis of the Segmentation and Classification of Pneumonia from Chest X-Ray Images", ICCSA-2021, Acharya Institute of Technology, Bengaluru.

Academic Reviewer

International Journal of Computer Vision; Computers in Biology and Medicine; Information Sciences; Machine Learning: Science and Technology; Applied Soft Computing Journal; Physics in Medicine and Biology; Indonesian Journal of Electrical Engineering and Computer Science; Digital Health; Inteligencia Artificial; Systems and Soft Computing; Medical Research Archives.

Workshops, FDPs, Resource Person Roles and Training

- 2026: Invited Keynote Speaker, 15th International Conference on Recent Challenges in Engineering and Technology (ICRCET 2026), Bangalore, India; keynote session listed in the official conference program and speaker page.
- 2025: FDP on "AI in Medical Imaging and Diagnostics: Current Trends and Challenges", E&ICT Academy, NIT Patna with IIITDM Jabalpur, IIT Guwahati, MNIT Jaipur, IIT Kanpur, IIT Roorkee, supported by MeitY.
- 2025: GIAN FDP on "Medical Informatics, Radiomics, and Image Analysis for Computer Aided Diagnosis", NIT Karnataka, Surathkal; presided by Prof. Rangaraj M. Rangayyan, University of Calgary.
- 2024: Resource Person, National Level Workshop on Blockchain Technology and Generative AI Models, Ghousia College of Engineering, Ramanagara.
- 2023: Resource Person, Two-Day National Workshop on Android Application Development using Java, Maharaja Institute of Technology Mysore.
- 2022: Resource Person, Month Internship Program on Deep Learning and its Applications, Hirasugar Institute of Technology; and Three-Day Workshop on Mobile App Development in Android, MIT Thandavapura.
- 2021: Resource Person, Two-Day Workshop on Deep Learning, St. Joseph's University; One-Week Workshop on Research Methodology, MIT Mysore; Letter of Appreciation from Bangalore Institute of Technology for Webinar on Applications of AI.
- 2019-2013: Resource person / participant in workshops and FDPs on research writing, machine learning, deep learning, data science, data mining, and lateral thinking across India and Nigeria.

Merits and Awards

- 2026: Sports Coordinator and Winner, MITM Mahadasara Sports Utsava 2026.
- 2026: Phase 1 Winner / Final-Round Shortlisted Team, AIISH Hack'A'Comm 2026.
- 2025: Winner, AIISH Hackathon on Assistive Technology.
- 2024: Best Paper Award, ADCIS-2024, BITS Pilani Goa; Best Paper Award, ERCICAM-2024, NITTE Meenakshi Institute of Technology / IEEE Bengaluru.
- 2021: Best Paper Award, ICCSA-2021, Acharya Institute of Technology, Bengaluru.
- 2016: Gold Medallist and Best Outgoing Student in PG Science, Computer Science, St. Joseph's College (Autonomous), Bangalore.
- 2015-2016: Management Scholarship, St. Joseph's College (Autonomous), Bangalore.

Languages, Technical Skills and Interests

Languages: English; Kannada (Basic). **Programming:** Python; Java; JavaScript; PHP. **Databases:** MySQL; PostgreSQL; Oracle. **Technical Skills:** Machine Learning; Deep Learning; Computer Vision; Medical Image Analysis; Explainable AI; Data Visualization; Research Methodology; Mobile Application Development; Big Data Analytics; DBMS. **Interests:** Athletics; Cooking; Cycling; Nature; Swimming; Travelling; Student Mentorship; Community-Oriented AI Innovation; Assistive Technology; Sports Coordination.

Academic Referees

Dr. Smitha Joyce Pinto, Associate Professor, Department of ECE, MIT Mysore, Karnataka - 571477, India; Email: `smithapinto_ece@mitmysore.in`.
Dr. Hemanth S.R., Associate Professor and Head, Department of CSE (Artificial Intelligence), MIT Mysore, Karnataka - 571477, India; Email: `hemanthsr_cse@mitmysore.in`, `hodai@mitmysore.in`.
Dr. Raghavendra K., Associate Professor, Department of CSE (Artificial Intelligence), MIT Mysore, Karnataka - 571477, India; Email: `raghavendrak_ai@mitmysore.in`.